

# THOUFIQ YAHYA

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## Senior Embedded Software Engineer | Automotive HMI

### EXECUTIVE SUMMARY

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Software Architect and Technical Lead with 13+ years of experience designing embedded software, automotive infotainment platforms, graphics systems and Human Machine Interfaces. Extensive experience spanning C++, Qt/QML, Embedded Linux, OpenGL ES, GPU programming, conversational AI, Android Automotive and software architecture. Proven track record leading cross-functional engineering teams and delivering production software for Mercedes-Benz MBUX and MB.OS platforms. Experienced across the complete software lifecycle, from research prototypes and framework development to production deployment, graphics optimization and technical leadership.

### FUNCTIONAL EXPERTISE

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#### Programming Languages

- C++
- Python
- JavaScript
- Rust
- Qt/QML
- Bash
- Kotlin

#### Graphics & Rendering

- OpenGL ES
- Shader Development
- Unity3D
- GLSL, HLSL
- Qt Rendering Framework
- Blender

#### Automotive

- Automotive IVI
- Voice User Interface (VUI)
- Human Machine Interface (HMI)
- Android Automotive OS

#### Embedded Systems

- Embedded Linux
- IPC (DBus)
- GCC Toolchain
- Docker
- CMake
- System Integration

#### Software Engineering

- Software Architecture
- SOLID Principles
- OOP
- TDD
- GMock
- UML
- Design Patterns
- CI/CD
- GTest

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Xpheno Staffing - Client MBRDI

December 2024 - Present

- Took ownership of an in-house 3D visualization engine after the project transitioned from another engineering team, ensuring continuity, maintainability, and feature development.
- Developed and enhanced a real-time automotive visualization engine built on Qt, responsible for rendering interactive 3D vehicle models used to demonstrate vehicle features and user interactions within Mercedes-Benz infotainment systems.
- Worked extensively with the Qt Graphics and Rendering Framework, gaining deep understanding of Qt's rendering pipeline, scene graph, and graphics architecture to implement scalable rendering solutions.
- Designed and implemented new visualization features using GLSL shader programming, developing GPU-accelerated rendering effects and custom visualizations for next-generation automotive HMI use cases.
- Optimized rendering performance, shader execution, and graphics workflows for resource-constrained embedded automotive hardware.
- Collaborated closely with UX designers, graphics engineers, and platform teams to integrate new rendering capabilities into production software.
- Contributed to the evaluation and migration strategy toward Unity-based rendering architecture, exploring integration approaches and next-generation graphics technologies for future Software Defined Vehicle (SDV) platforms.
- Participated in architectural discussions involving graphics engine evolution, rendering pipelines, asset workflows, and long-term technology direction.

#### **Mercedes-Benz Research & Development India**

**April 2016 - September 2023 (7.5yrs)**

- Led architecture and development of next-generation production widgets for Mercedes-Benz MBUX and MB.OS infotainment platforms.
- Designed reusable UI frameworks using modern C++, Qt6 and QML.
- Built advanced UI controls exceeding standard Qt Quick Controls capabilities, including animated buttons, progress indicators, shader-driven effects and complex interaction patterns.
- Designed reusable component architecture adopted across multiple infotainment applications.
- Worked closely with UX designers to translate high-fidelity concepts into production-quality embedded software.
- Led code reviews, architecture discussions and technical mentoring for a team of engineers.
- Collaborated with platform, CI/CD and integration teams across Germany and India.
- Designed and developed a prototype in-house Sound Dialogue System (SDS) for Mercedes-Benz to evaluate alternatives to third-party voice assistant vendors.
- Built a context-aware dialogue manager capable of dynamically switching between multiple Natural Language Understanding (NLU) providers.
- Integrated multiple cloud-based NLU engines including Dialogflow, Rasa, Wit.ai and Houndify.
- Developed a modular architecture enabling runtime selection of the most suitable NLU engine based on user intent.
- Demonstrated feasibility of an internally developed conversational AI platform for future automotive voice assistants.

#### **L & Technology Services - Technical Consultant**

**June 2014 - April 2016 (2yrs)**

- Worked as a consultant for Cisco NDS on embedded television middleware and Electronic Program Guide (EPG) development.
- Developed embedded user interfaces using Adobe Flash, ActionScript and later Qt/QML technologies.
- Participated in the organization's migration from Flash-based UI frameworks to modern Qt/QML-based embedded applications.
- Collaborated with Linux platform teams on embedded software integration.
- Optimized UI performance and memory usage for resource-constrained embedded devices.
- Gained extensive experience in Embedded Linux, C++, Qt/QML and embedded application architecture.

#### **I2Play Interactive - Game Developer**

**Nov 2012 – June 2014 (1.5 yrs)**

- Developed multiplayer educational games focused on cognitive learning and problem-solving for children.
- Designed gameplay systems including puzzles, connect-the-dots and interactive learning mechanics.
- Built Flash-based multiplayer games using ActionScript.
- Integrated SmartFoxServer to support real-time multiplayer gameplay.
- Developed backend services using PHP and MySQL.
- Collaborated with artists and designers to build engaging educational experiences.
- Optimized rendering performance and game responsiveness across browsers.

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**Excel Goodies - Game Developer / Web Designer****June 2011 – Jan 2012 (6 months)**

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- Developed interactive educational games and rich internet applications using Adobe Flash and ActionScript.
- Designed and implemented responsive web applications using PHP, HTML, XHTML, JavaScript and CSS.
- Created UI mockups, artwork, game assets and branding materials using Photoshop, Illustrator and CorelDRAW.
- Worked across both design and development, translating visual concepts into production-ready applications.
- Collaborated directly with clients in a startup environment, delivering complete end-to-end web and multimedia solutions.

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**AWARDS & RECOGNITION**

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- Star Performer — L&T Technology Services (2015)
  - Winner, Innovation Boot Camp — Mercedes-Benz R&D (2019)
  - Silver Star — Mercedes-Benz R&D (2020)
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**EDUCATION**

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**Bachelor of Information Technology****2006 - 2009**